上海科技大学

2021-2022 学年第二学期本科生期末考试卷

开课单位：物质学院

授课教师：李涛

考试：普通化学 II

课程代码：CHEM1111.01

姓名：.....

学号：.....

学院：.....

班级：

考生须知：

1. 请严格遵守考场纪律，禁止任何形式的作弊行为。
2. 参加闭卷考试的考生，除携带必要考试用具外，书籍、笔记、掌上电脑和其他电子设备等物品一律按要求放在指定位置。
3. 参加开卷考试的考生，可携带教师指定的材料独立完成考试，但不准相互讨论，不准交换材料。

考试成绩录入表：

题目	1	2	3	4	5	6	**总分**
计分							
复核							

评卷人签名： **日期：** **复核人签名：** **日期：**

1																	18
1 **H** 1.008	2											13	14	15	16	17	2 **He** 4.0026
3 **Li** 6.94	4 **Be** 9.0122											5 **B** 10.81	6 **C** 12.011	7 **N** 14.007	8 **O** 15.999	9 **F** 18.998	10 **Ne** 20.180
11 **Na** 22.990	12 **Mg** 24.305	3	4	5	6	7	8	9	10	11	12	13 **Al** 26.982	14 **Si** 28.085	15 **P** 30.974	16 **S** 32.06	17 **Cl** 35.45	18 **Ar** 39.948
19 **K** 39.098	20 **Ca** 40.078	21 **Sc** 44.956	22 **Ti** 47.867	23 **V** 50.942	24 **Cr** 51.996	25 **Mn** 54.938	26 **Fe** 55.845	27 **Co** 58.933	28 **Ni** 58.693	29 **Cu** 63.546	30 **Zn** 65.38	31 **Ga** 69.723	32 **Ge** 72.630	33 **As** 74.922	34 **Se** 78.97	35 **Br** 79.904	36 **Kr** 83.798
37 **Rb** 85.468	38 **Sr** 87.62	39 **Y** 88.906	40 **Zr** 91.224	41 **Nb** 92.906	42 **Mo** 95.95	43 **Tc** (98)	44 **Ru** 101.07	45 **Rh** 102.91	46 **Pd** 106.42	47 **Ag** 107.87	48 **Cd** 112.41	49 **In** 114.82	50 **Sn** 118.71	51 **Sb** 121.76	52 **Te** 127.60	53 **I** 126.90	54 **Xe** 131.29
55 **Cs** 132.91	56 **Ba** 137.33	57-71 *	72 **Hf** 178.49	73 **Ta** 180.95	74 **W** 183.84	75 **Re** 186.21	76 **Os** 190.23	77 **Ir** 192.22	78 **Pt** 195.08	79 **Au** 196.97	80 **Hg** 200.59	81 **Tl** 204.38	82 **Pb** 207.2	83 **Bi** 208.98	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
87 **Fr** (223)	88 **Ra** (226)	89-103 #	104 **Rf** (265)	105 **Db** (268)	106 **Sg** (271)	107 **Bh** (270)	108 **Hs** (277)	109 **Mt** (276)	110 **Ds** (281)	111 **Rg** (280)	112 **Cn** (285)	113 **Nh** (286)	114 **Fl** (289)	115 **Mc** (289)	116 **Lv** (293)	117 **Ts** (294)	118 **Og** (294)

* Lanthanide series	57 **La** 138.91	58 **Ce** 140.12	59 **Pr** 140.91	60 **Nd** 144.24	61 **Pm** (145)	62 **Sm** 150.36	63 **Eu** 151.96	64 **Gd** 157.25	65 **Tb** 158.93	66 **Dy** 162.50	67 **Ho** 164.93	68 **Er** 167.26	69 **Tm** 168.93	70 **Yb** 173.05	71 **Lu** 174.97
# Actinide series	89 **Ac** (227)	90 **Th** 232.04	91 **Pa** 231.04	92 **U** 238.03	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (262)

Physical Constants

Avogadro's number	$N_A = 6.02214179 \times 10^{23}\ \text{mol}^{-1}$
Bohr radius	$a_0 = 0.52917720859\ \text{Å} = 5.2917720859 \times 10^{-11}\ \text{m}$
Boltzmann's constant	$k_B = 1.3806504 \times 10^{-23}\ \text{J K}^{-1}$
Electron charge	$e = 1.602176487 \times 10^{-19}\ \text{C}$
Faraday constant	$F = 96{,}485.3399\ \text{C mol}^{-1}$
Masses of fundamental particles:	
Electron	$m_e = 9.10938215 \times 10^{-31}\ \text{kg}$
Proton	$m_p = 1.672621637 \times 10^{-27}\ \text{kg}$
Neutron	$m_n = 1.674927211 \times 10^{-27}\ \text{kg}$
Permittivity of vacuum	$\epsilon_0 = 8.854187817 \times 10^{-12}\ \text{C}^{-2}\ \text{J}^{-1}\ \text{m}^{-1}$
Planck's constant	$h = 6.62606896 \times 10^{-34}\ \text{J s}$
Ratio of proton mass to electron mass	$m_p/m_e = 1836.15267247$
Speed of light in a vacuum	$c = 2.99792458 \times 10^{8}\ \text{m s}^{-1}$ (exactly)
Standard acceleration of terrestrial gravity	$g = 9.80665\ \text{m s}^{-2}$ (exactly)
Universal gas constant	$R = 8.314472\ \text{J mol}^{-1}\ \text{K}^{-1}$ $= 0.0820574\ \text{L atm mol}^{-1}\ \text{K}^{-1}$

Values are taken from the 2006 CODATA recommended values, as listed by the National Institute of Standards and Technology.

Problem 1. (Total 16 pts)

The normal boiling point (at 1 atm) of liquid N_2 is 77.35 K; the enthalpy of vaporization (ΔH_{vap}) at that temperature is 5.57 kJ•mol^{-1}. The heat capacity of gaseous N_2 at constant pressure is 29.12 J•mol^{-1}•K^{-1}.

(a) (2 pts) How much energy does it take to vaporize 1.00 kg of liquid N_2 at 1 atm? (Temperature is maintained at 77.35 K)

(b) (2 pts) What is the percentage of volume expansion (膨胀) of gaseous N_2 from 77.35 K to 298.0 K under constant pressure?

(c) (8 pts) Calculate q, w, ΔH, and ΔU for the following change in state:
3.00 mol N_2 (l, 1 atm, 77.35 K) → 3.00 mol N_2 (g, 1 atm, 298.0 K)
Assume that the gas behaves ideally, and that the volume occupied by the liquid is negligible (可忽略的).

(d) (4 pts) Calculate the entropy change from 3.00 mol liquid N_2 (77.35 K, 1 atm) to 3.00 mol gaseous N_2 (298 K, 1 atm). Assuming it is a reversible process.

Problem 2. (18 pts)

Cd^{2+} can react with Cl^- to form a complex ion $CdCl_4^{2-}{}_{(aq)}$. The formation constant K_f = 8.00 × 10^2. On the other hand, CdS is an insoluble compound with a dissolution equilibrium constant $K^\dagger$ of 7.00 × 10^{-27}. The acid dissociation constants of H_2S are 7.00 (pK_{a1}) and 12.00 (pK_{a2}).

(a) (2 pts) Write down a balanced ionic equation of CdS reacting with HCl.

(b) (2 pts) What is the expression of the dissolution equilibrium constant ($K^\dagger$) of CdS?

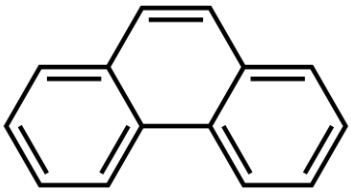
(c) (4 pts) What is the solubility (in g/L) of CdS in pure water?

(d) (4 pts) What is the solubility (in g/L) of CdS in a 5.00 M KCl solution?

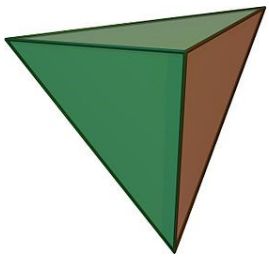
(e) (6 pts) What is the solubility (in g/L) of CdS in a 6.00 M HCl solution? What are the concentrations of [HS^-] and [S^{2-}] after reaching dissolution equilibrium?

Problem 3. (15 pts)

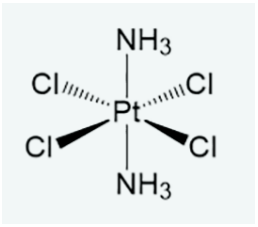
(a) (9 pts) List all the symmetry operations (对称操作) for the following objects (For example: 3×σ_v ; 1×σ_h; 2×C_4; 4×C_2; 1×i).



i

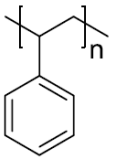


ii (tetrahedron, 正四面体)

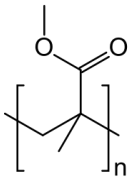


iii (Consider NH_3 as a spherical molecule)

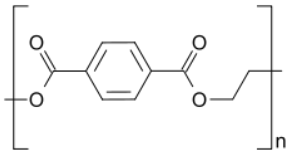
(b) (6 pts) Draw the structures of the corresponding monomers of the following polymers



polystyrene



poly(methyl methacrylate)



PET

Problem 4. (Total 17 pts)

(a) (4 pts) List 4 applications (应用) of zeolites

(b) (4 pts) Correlate (关联) the following nanomaterials with their special properties

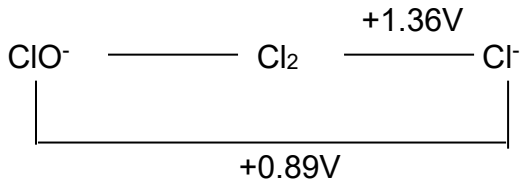
A. DNA origami (折纸)	i. As a conductive coating (导电涂层)
B. Metallic nanoparticles	ii. Self-assemble into complex (复杂) shapes
C. Graphene	iii. For biomolecule detection (识别)
	iv. Localized surface plasma resonance (LSPR)

(c) (5 pts) Assuming (假设) a polymer sample contains the following molecules, calculate the number average molecular weight (M_n), weight average molecular weight (M_w) and the polydispersity index (PDI).

Molecular weight (g/mol)	# of molecules
8300	2
14000	3
19500	5
26000	4
33000	1

(d) (4 pts) What is the difference between chain-growth polymerization and step-growth polymerization?

Problem 5. (Total 18 pts)



(a) (4pts) Write down the two half reactions for ClO^- reduced to Cl^- (occurs in basic condition) and Cl_2 to Cl^-.

(b) (4pts) Based on the standard reduction potential values given above, calculate $E^o_{ClO^-/Cl_2}$.

(c) (3pts) Predict if Cl_2 can disproportionate into ClO^- and Cl^- (STP condition) and provide reasoning.

(d) (3 pts) What is the equilibrium constant of the reduction of ClO^- to Cl^- at 25 ^oC?

(e) (4 pts) Consider a buffer solution containing 0.100 M Fe^{2+}. The pH is maintained at 11. What is the minimum concentration of NaClO required to oxidize Fe^{2+} to Fe^{3+}? ($E^o_{Fe^{3+}/Fe^{2+}}$ = 0.770 V)

Problem 6. (Total 16 pts)

The rate constant of the first-order reaction

$$2N_2O(g) \longrightarrow 2N_2(g) + O_2(g)$$

is 0.75 s^{-1} at 1000 K and 0.87 s^{-1} at 1030 K

(a) (3 pts) Calculate the activation energy (E_a) of the reaction

(b) (3 pts) Calculate the pre-exponential factor (A, from the Arrhrenius equation) of the reaction.

(c) (4 pts) What would be the predicted rate constant at 1100K?

(d) (6 pts) In a sealed reaction vessel, the initial partial pressure of N_2O is 5 atm. What is the half-life time of the reaction? How long will it take to have the partial pressure of N_2O decrease to 0.1 atm?